

Farhan Ali

Medical Laboratory Technologist

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SUMMARY

Research-oriented Medical Laboratory Technology graduate with a strong academic background in biomedical sciences and a keen interest in molecular biology and genetics. Experienced in research methodology, scientific literature analysis, and data interpretation. Eager to contribute to innovative research addressing the molecular basis of human diseases and advancing precision medicine.

EDUCATION

Khyber Medical University (KMU), Institute of Paramedical Sciences, (IPMS) 2021 – 2025 | Peshawar
Bachelor of Science in Medical Laboratory Technology (BS MLT)

CGPA : 3.69/4.0

Academic Standing: Ranked among top 6 students of the class

Government Higher Secondary School Sakhakot 2019 – 2021 | Malakand
Higher Secondary School Certificate (F.Sc Pre-Medical)

Marks: 960 / 1100

Academic Standing: College 1st Position Holder

ACADEMIC ACHIEVEMENTS

Thesis Defense Commendation, KMU Institute of Paramedical Sciences, IPMS Peshawar 2026
Successfully defended undergraduate research thesis with formal appreciation and high praise from external academic examiners.

Academic Excellence (4.0 GPA), KMU Institute of Paramedical Sciences, IPMS Peshawar 2025
Achieved a perfect 4.0 GPA across the final two academic semesters (7th and 8th), driven by exceptional performance in advanced research methodology coursework and the rigorous execution of an independent clinical research thesis.

Prime Minister's Laptop Scheme (Phase IV) –Awarded on academic merit 2025
Government Of Pakistan
Awarded a highly competitive, merit-based technology grant in recognition of an outstanding cumulative GPA, reflecting distinguished expertise and academic excellence in core laboratory sciences and research-oriented coursework.

Board Topper / Early Education Honors – Stoori Da Pakhtunkhwa Program. 2021
Board of Intermediate and Secondary Education Malakand.
Honored by the Elementary and Secondary Education Department for exceptional academic performance and provincial-level ranking.

RESEARCH EXPERIENCE / ACADEMIC PROJECTS

UNDERGRADUATE RESEARCH

2025 – 2026 | Pakistan

Institute of Paramedical Sciences– Khyber Medical University Peshawar

Thesis: *Psychosocial Challenges and Quality of Life in Children with Transfusion Dependent Beta Thalassemia and Associated Burden on Their Parents/Guardians*

Supervisor: Mr. Awal Mir, Lecturer, Department of Medical Laboratory Technology

- **Methodology & Clinical Assessment:** Designed and executed a descriptive cross-sectional study, recruiting a clinical sample of 200 child-parent dyads at the Peshawar Institute of Medical Sciences (PIMS). Administered structured bilingual (English/Urdu) interviews utilizing validated tools, including the PedsQL™ 4.0 Generic Core Scale and the Zarit Burden Interview.
- **Biostatistical Analysis:** Processed complex clinical data using SPSS (Version 27) and Microsoft Excel; executed advanced data analysis scripts within **R Studio**. Performed independent samples t-tests, one-way ANOVA, and Pearson correlation analyses to evaluate complex interrelationships between child distress and family strain.
- **Research Outcomes:** Defended thesis with formal academic commendation. Revealed strong negative correlations between psychosocial difficulties and quality of life, establishing local evidence to advocate for multidisciplinary thalassemia care models.
- **Submitted Manuscript:** Mir, A., Pervaiz, H., Rehman, A., Hamza, M., **Ali, F.**, Muhammad, J., & Khan, A. A. (2026). *Psychosocial difficulties, Health-Related Quality of life, And Caregiver Burden in Children with Transfusion-Dependent B-Thalassemia*. Manuscript submitted for publication to *Thalassemia Reports*.

COMPUTATIONAL RESEARCH PROJECTS

Clinical NGS Variant Annotator & Diagnostic Pipeline

08/2026

Independent Bioinformatics Project | Hosted on GitHub: github.com/farhanali-researcher

- **Engineered** an automated Python-based bioinformatics pipeline to ingest targeted Next-Generation Sequencing (NGS) variant panels and execute high-throughput genomic annotation.
- **Integrated** the Ensembl REST API using dynamic web requests to query international genomic databases and filter out benign background single-nucleotide polymorphisms (SNPs).
- **Automated** clinical phenotype extraction and pathogenicity screening, successfully isolating and mapping disease-causing variants to severe hematological disorders, including Beta-Thalassemia and Sickle Cell Disease.
- **Optimized** data processing workflows with robust error-handling and server-side rate-limiting protocols to ensure stable, scalable execution during batch biomedical data-mining.

Genomic Sequence Analysis & Annotation Pipeline

07/2026

Independent Bioinformatics Project | Hosted on GitHub: github.com/farhanali-researcher

- **Engineered** an automated bioinformatics pipeline using Python and the Biopython framework for advanced genomic sequence analysis and molecular characterization.
- **Integrated** BLAST algorithms with NCBI genomic databases to execute precise gene-finding, sequence retrieval, and structural stability analysis.
- **Translated** complex biological datasets into robust R scripts (ggplot2), ensuring workflow reproducibility and generating publication-ready statistical data visualizations.

RESEARCH INTERESTS

- **Translational Hematology & Molecular Diagnostics:** Bridging clinical pathology with genetic data to improve the diagnosis and management of chronic hemoglobinopathies, specifically Transfusion Dependent Beta Thalassemia.
- **Computational Biology & Bioinformatics:** Utilizing genetic sequence analysis and computational tools (e.g., BLAST) for molecular characterization and gene finding.

- **Clinical Epidemiology & Biopsychosocial Health:** Evaluating Health-Related Quality of Life (HRQoL), caregiver burden, and the psychosocial determinants of chronic pediatric diseases.
- **Advanced Laboratory Sciences:** Optimizing diagnostic workflows, quality control, and clinical data analysis to improve patient outcomes in resource-limited healthcare settings.

CLINICAL EXPERIENCE

Medical Laboratory Technologist (Internship) 2025 – 2026 | Peshawar, Pakistan
City Medical Welfare Laboratory

- **Rapid-Response Diagnostics:** Managed high-throughput specimen processing and diagnostic testing in a fast-paced laboratory directly supporting emergency and outpatient cases.
- **Clinical Pathology & Analysis:** Executed routine and urgent assays across hematology, biochemistry, and serology to facilitate immediate, data-driven clinical decision-making.
- **Quality & Specimen Management:** Conducted precise phlebotomy and maintained strict chain-of-custody protocols for clinical samples while ensuring rigorous adherence to standard operating procedures (SOPs).

Medical Laboratory Technologist (Internship) 2024 – 2025 | Peshawar, Pakistan
Lady Reading Hospital (LRH)

Rotated through all core diagnostic departments in a premier tertiary care hospital. Managed high-throughput workflows by processing **80+ clinical specimens** per shift in each section while strictly following QA/QC protocols.

- **Hematology:** CBC, smear examination, ESR, coagulation profiles, Hb electrophoresis
- **Microbiology:** culture techniques, staining, biochemical testing
- **Serology:** ELISA, Coombs' test
- **Clinical Chemistry:** routine & special chemistry on automated analyzers
- **Molecular Biology:** PCR-based diagnostic exposure
- **Histopathology:** tissue processing and slide preparation
- **Blood Bank:** grouping, cross-matching
- **Phlebotomy:** venipuncture and arterial sampling

TECHNICAL SKILLS

- **Biostatistics & Programming:** R Studio (statistical analysis scripts, data visualization), Python (Biopython framework), Linux/Bash Scripting, SPSS v27 (One-way ANOVA, independent t-tests, Pearson correlation).
- **Bioinformatics & Sequence Analysis:** Sequence alignment (BLAST, NCBI), gene-finding tools, genome annotation, structural bioinformatics, and 3D protein modeling (AlphaFold, SWISS-MODEL), genomic/proteomic databases (UniProt, PDB, ExPASy).
- **Clinical Diagnostics (Wet Lab):** PCR-based molecular diagnostics, Hematology (CBC, Hb electrophoresis), Serology (ELISA, Coombs' testing), Blood Banking (cross-matching), comprehensive microbiological cultures, and stringent QA/QC protocol management.
- **Scientific Tools & Research Methodology:** Reference Management (Mendeley), cross-sectional study design, clinical data collection protocols, scientific writing, and Canva (for scientific poster design and presentations).

CONFERENCES, SEMINARS & WEBINARS

DNA Extraction from Biopsy Samples & Gene Amplification Workshop 
BJ Micro Lab, July 2026

Online Course: How to Write a Review Paper

Organized by Collab & Rise Global Society, June 2026.

"Building your portfolio with Github" Session

Hosted by Global Research Innovation Network during June Skill Development June, 2026.

Hands-on Session: Introduction to Bioinformatics; Tools, Databases, and Applications in Modern Biology

conducted by ScientificHubX, April 2026.

Educational Webinar: Unlocking Albumin Creatinine Ratio (UACR)

Hosted by Erba Mannheim and Transasia, April 2026.

Aspire Leadership Awareness Program: Unlocking Leadership Potential

Collaboratively organized by Aspire Institute, NYLP, & AFPGMI (NUMS), April 2026.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Participant, NORD Students for Rare

08/2026

National Organization for Rare Disorders (NORD) Aug 2026 – Present

- Engaged in rare disease advocacy and education initiatives to support patients and advance awareness of genetic disorders within the medical and scientific community.

Participant, Aspire Leadership Program (Cohort 3)

07/2026

Aspire Institute (Founded by Harvard University Faculty) July 2026 – Present

- Selected for a globally competitive, fully funded leadership development program designed for first-generation college students and early-career professionals.
- Engaging in rigorous professional development, cross-cultural collaboration, and leadership training to address complex global challenges.
- Participating in interactive case study discussions and masterclasses led by world-class educators and industry leaders.

CERTIFICATIONS & TRAINING

Bioinformatics for Biologists: An Introduction to Linux, Bash Scripting, and R

08/2026 | Online

By Wellcome Connecting Science via FutureLearn

Unleash Your Potential: AI Fundamentals , University of Bristol via FutureLearn

08/2026 | Online

Digital PCR 101 , Bio-Rad Academy

08/2026 | Online

Writing in the Sciences , from Stanford University, Coursera

07/2026 | Online

Good Clinical Practice , from NIDA Clinical Trials Network

06/2026 | Online

PCR Skills Development

11/2025 | Peshawar, Pakistan

Hands on Workshop at Khyber Medical University Peshawar

Laboratory Biosafety & Biosecurity

11/2025 | Peshawar, Pakistan

Hands on Workshop at Khyber Medical University Peshawar

PROFESSIONAL AFFILIATIONS

Association of Khyber Medical University Alumni (AKmUA)

American Society for Clinical Pathology (ASCP)

American Society for Microbiology (ASM)

American Association for Cancer Research (AACR)

Association for Molecular Pathology (AMP)

LANGUAGES

- **English:** Full Professional Proficiency (CEFR C1)
- **Urdu:** Native / Bilingual Proficiency
- **Pashto:** Native / Bilingual Proficiency

REFERENCES

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